Validate Three Nodes

AlgoExpert · Hard · Binary Search Trees

The problem

- Is nodeTwo a descendant of one of nodeOne/nodeThree and an ancestor of the other?

Intuition

- Node two must sit on the path between the other two, so search down from node one and node three.

Approach

- 1. Check if node two descends from node one or node three
- 2. Then check that the other one descends from node two
- 3. Use BST search for each step

Java solution

```java
public class ValidateThreeNodes {
    public static boolean validateThreeNodes(BST one, BST two, BST three) {
        if (descends(two, one)) return descends(three, two);
        if (descends(two, three)) return descends(one, two);
        return false;
    }

    /** true if `target` is reachable going down from `from`. */
    private static boolean descends(BST from, BST target) {
        BST c = from;
        while (c != null && c != target) c = target.value < c.value ? c.left : c.right;
        return c == target;
    }
}
```

Complexity

- Time: O(h)
- Space: O(1)

Recap

- Node two must sit on the path between the other two, so search down from node one and node three.
- Complexity: O(h)